

Accumulated risk of body postures in assembly line balancing problem and modeling through a multi-criteria fuzzy-genetic algorithm

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ABSTRACT

Monotonous body postures during repetitive jobs negatively affect assembly-line workers with the developing of Work-related Musculoskeletal Disorders (WMSDs). Ergonomics specialists have offered auxiliary posture diversity to deal with the lack of varieties, especially for high-risk ones. Meanwhile, Assembly Line Balancing (ALB) problem has been recognized as a prior thinking to (re)configure assembly lines via the balancing of their tasks among their workstations. Some conventional criteria, cycle time and overall workload are often considered during the balancing. This paper presents a novel model of ALB problem that incorporates assembly worker postures into the balancing. In addition to the conventional ALB criteria, a new criterion of posture diversity is defined and contributes to enhance the model. The proposed model suggests configurations of assembly lines via the balancing; so that the assigned workers encounter the opportunities of changing their body postures, regularly. To address uncertainties in the conventional criteria, a fuzzy goal programming is used, and an appropriate genetic algorithm is developed to deal with the model. Various computational tests are performed on the different models made with combinations of the three criteria mentioned above. Comparing the pay-offs among the combinations, results show that well balanced task allocation can be obtained through the proposed model.

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1. Introduction

Assembly Line (AL) is a manufacturing process through which interchangeable parts are sequentially added to a product at workstations to produce a final product in due time. To design an assembly line, Assembly Line Balancing (ALB) problem is initially defined to deal with allocating assembly tasks among the assembly workstations considering some predefined criteria (Becker & Scholl, 2006). Processing time (called cycle time), the number of workstations, workload smoothness, work relatedness and smoothness/relatedness have been mostly considered as primary criteria in the balancing problem so far (Tasan & Tunali, 2008). Since ALB problem is a long-term decision and might demand capital investments, it is critical to consider further possible problems resulting from many factors (Gen, Cheng, et al., 2008).

Work-related Musculoskeletal Disorders (WMSDs) have been recognized as a health issue in most industries (Mattila, Karwowski, & Vilkkki, 1993). Besides that, the problems may also lead to considerable productivity loss among workers with highly physical jobs (Meerding, Ijzelenberg, Koopmanschap, Severens, & Burdorf,

2005). Meanwhile, posture, repetition and duration have been identified as the main risk factors of WMSDs in highly repetitive tasks (including assembly tasks) (Bridger, 2003). So then, some researchers have recently employed the mentioned risk factors into ALB problem, and they developed the models regarding the new criteria (e.g. Choi, 2009; Otto & Scholl, 2010).

The purpose of this research is to benefit from posture diversity concept into ALB problem to decline the risk of workers' postures. Despite many ALB models using the risk factors, no one has particularly focused on the body postures of assembly workers with high-risk posture demanding during the tasks. Even with the balanced processing time and (overall) physical workload, the resulting pressures are locally imposed on individual parts of workers' body. It is due to the nature of each assembly task that demands specified body postures. Actually, keeping assembly workers in monotonous postures might cause fatigue earlier, and it is often assumed that controlling of fatigue can also control WMSDs (Ford, Notestine, & Hill, 2000). Therefore, with the proposing ALB model, each worker meets set of postures not to be in continuously dangerous or high-risk posture as possible as it can.

To contribute the risks of postures, measure or scale systems are obviously required. There are five general types of workload measurements: comparison, decision tree, sub-scale sets, single rating scale and task-analysis (Gawron, 2008). Rating methods based on single or multiple (sub) scales are more popular among ergonomics

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specialists and range from relatively simple to complex. For example, Kee and Karwowski (2003) have introduced a ranking system to evaluate joint motion stressful, and later they developed a scaling measuring for joint angles among females based on static joint postures (Kee & Karwowski, 2004).

Many physical exposure assessments emphasize posture-based methods; and these methods evaluate potential risks associated with postures that occur during working hours (Li & Buckle, 1999). Quick exposure check (QEC), rapid entire body assessment (REBA), rapid upper-limb assessment (RULA), postural loading on the upper-body assessment (LUBA) are samples of the many methods that have been reviewed by Takala, et al. (2010). In this paper, we employ a simplified version of OWAS – Ovako Work Assessment System – (Karhu, Kansu, & Kuorinka, 1977) with a five-point score for each body part (back, leg, neck, etc.) to determine the risk levels for postures during each task. The formulations are discussed in the following section.

Most these assessments deal with human observation and/or approximation. So then, linguistic variables, Fuzzy sets and neural networks have been broadly used to improve these measurement techniques. For instance, Chen, Jung, and Peacock (1994) developed a model with Analytic Hierarchy Process (AHP), which measures workload through paired comparisons and the concept of fuzzy set to predict ergonomic workload stress. Karwowski, Lee, Marras, and Rodrick (2003) have studied activities of trunk muscle during manual lifting through a neuro-fuzzy model, and later, Karwowski et al. have used a fuzzy relational rule network to model of trunk muscle in lifting activity (Karwowski et al., 2006). Hou et al. have also used a recurrent fuzzy neural network to estimate the dynamic spinal force (Hou, Zurada, Karwowski, Marras, & Davis, 2007).

ALB problem is further characterized by uncertain task times, which mainly arise from individual variations in work rate, skill, motivation and learning and also the failure sensitivity of complex tasks (Buzacott, 1990; Robinson, McClain, & Thomas, 1990; Scholl & Becker, 2006). This uncertainty can be represented by fuzzy numbers to reduce error (Gen, Tsujimura, & Li, 1996). Besides that, no precise measurements are available for overall workload (physical workload) assessment, because all mentioned assessments have been based on subjective questionnaires and observational methods (Choi, 2009; Takala et al., 2010). Then a fuzzy mathematical ALB model, and a genetic algorithm based on three goals are proposed. Two fuzzy goals are minimizing the processing time and the overall physical workload that are considered individually from posture diversity in the line balancing.

The information included in this paper is organized as follows: the following section presents the problem statement through a simple example. Section 3 details the definition of accumulated risk of postures that is required for following mathematical formulations along fuzzy goals and criteria. An illustrative example is presented in this section. Section 4 presents genetic algorithm and the computational results from the example illustrated in Section 2 and other data sets are followed over Sections 5 and 6. Section 7 ends this paper with some conclusions and further lines.

2. Problem statement

Consider a small AL with two workstations and four tasks named by T_1 , T_2 , T_3 and T_4 . ALB constraints are usually presented by a precedence graph that shows the specified assembly tasks that must be performed earlier than some other tasks. To state the problem, here, it is not under consideration. Furthermore, suppose that three areas (of body) are only considered: A_1 (back), A_2 (arms) and A_3 (legs). And each task demands a specified body posture (at least a high-risk posture at the selected time) to be performed. The demanding postures for each task and the involving areas are

shown in Fig. 1a and b; and simplified OWAS posture code definitions (Mattila et al., 1993) are presented by Fig. 1c. The associated OWAS-related postural risk levels (RLs) for each task and on each area (of body) are calculated by looking up from the OWAS posture codes – see Fig. 1d. Having lower RL values for each workstation indicate that assigned worker will be treated with low-risk postures during his job. Moreover, the overall workloads for all tasks are assumed to be equal, and the approximated times bound for the tasks (approximated range of possible time to finish the task) are presented in Fig. 1d. It is applied to balance the tasks between Workstations 1 and 2 and to minimize the cycle time interval.

Consider three configurations shown in Table 1 can be implemented for the mentioned ALB problem without any constraint or precedence limitation. There are RL values of {2,2} on A_2 at Workstation 2 and {1,2} on A_1 at Workstation 1 in Configuration 1 and 3, respectively, Whereas Configuration 2 does not release such these sets of high-value RLs. Configuration 1 offers an assignment with the highest values of RLs (with two 2s) on A_2 at Workstation 2. So the workers performing the assigned tasks at Workstation 2 can experience continuous pressure on A_2 . However, the configuration provides a shortest maximal value of the total time intervals, the range of 65–73. In contrast to this configuration, Configuration 2 has a longest total time interval at workstation 2, a range of 75 and 85; that it makes the related cycle time longer for the entire of the assembly line. Configuration 3 grants the maximum total time interval in a range of 70 and 75 with the maximum pressure on A_1 , {1,2}.

Looking back at the introduction, the effort is to preclude the development of WMSDs through the balancing in the way that no worker faces continuous (monotonous) high-risk postures. The above example illustrates that this effort is not completely in harmony with other objectives like the minimization of the maximal total time interval (cycle time) in ALB problem. As it takes into account, the balancing becomes a compromise decision, and decision maker(s) should treat these conflict objectives to optimize with reasonable goals or range of values.

To contribute RLs to ALB problem, a new objective, accumulated risk of postures is defined and then formulated in the following sections. It simply approximates the considered diversity in the proposing ALB model.

3. Definition and mathematical formulations

3.1. Accumulated risk of postures

First of all, a parameter is required to estimate a level of high-risk posture diversity in an assembly line. The parameter depends on the considering areas of workers' body and the assigned tasks performing as their job. And its value should be calculated according to demanding postures during the assembly job. In fact, the value will be increased with the lack of capability of change in postures from high-risk to low-risk ones during the assembly job in one workstation.

Accumulated Risk of Postures (ARPs) is first defined for workstations. ARP value is calculated through the changes of risk level of postures in repetitive jobs that its benefits have been approved (by Mathiassen (2006), Mathiassen, Moller, and Forsman (2003), Mathiassen and Winkel (1991)). ARP value indicates the lack of high-risk posture changes during a cycle of performing tasks in workstations. To this end, the definition utilizes relatively high RL values that probably appear in workstations. Actually, workstations with low values in APR can be good options to prevent monotonous postures with high-risk during assembly jobs.

All efforts through the definition, is to make a new objective for ALB problems with posture diversity in their workstations. It

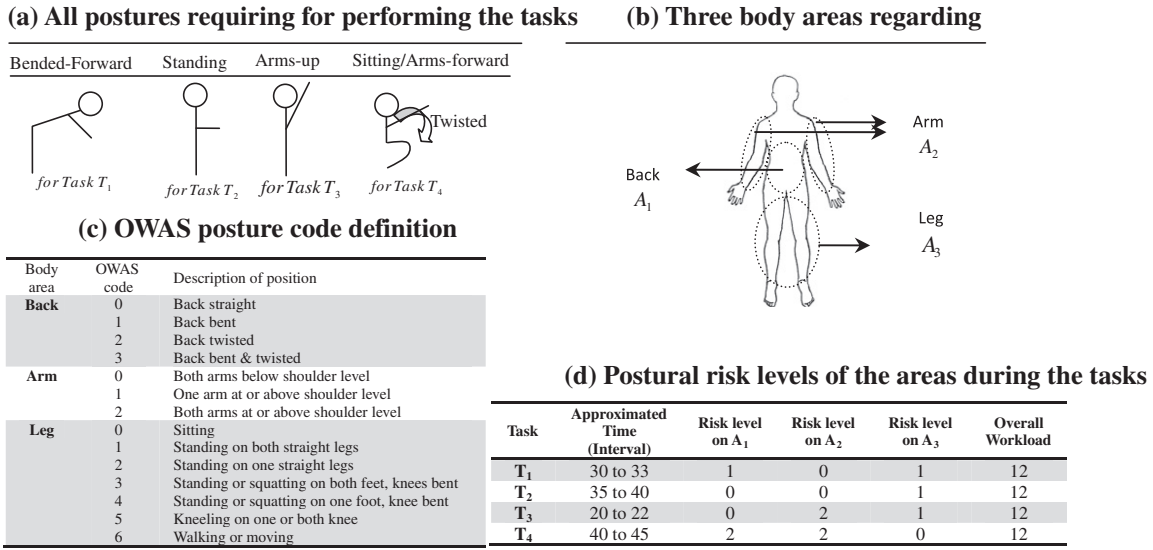


Fig. 1. An assembly line with four tasks and three body parts.

Table 1
Three proposed configurations for AL.

	WorkStation 1 T_1 and T_2	WorkStation 2 T_3 and T_4
Configuration 1		
Total time interval:	65–73	60–65
RLs on A_1 :	0, 1	0, 2
RLs on A_2 :	0, 0	2, 2
RLs on A_3 :	1, 1	1, 0
Configuration 2		
Total time interval:	50–55	75–85
RLs on A_1 :	0, 1	0, 2
RLs on A_2 :	0, 2	0, 2
RLs on A_3 :	1, 1	1, 0
Configuration 3		
Total time interval:	70–75	55–62
RLs on A_1 :	1, 2	0, 0
RLs on A_2 :	0, 2	0, 2
RLs on A_3 :	0, 1	1, 1

prevents to accumulate maximum risk of postures that often affect some areas of workers' body. According to the aforementioned expectations of ARP, a clear definition of ARP should be appropriated. Hence, a stepwise mathematical definition of ARP is considered to develop the definition for body areas in one workstation, first, and then for workstations and finally for the whole assembly line.

Let T_j represents an index set of assembly tasks assigned to Workstation j in the line; and $RL_i(l)$ shows RL value of Task l on Area i (RL has been already defined in Section 2). NW and m indicate the number of workstations and body areas considered in ALB problem, respectively. ARP_j^i related to Workstation j and Area i is defined as follows:

$$ARP_j^i = \sum_{(l, l' \in T_j) \text{ and } (i, i' \in I)} [((RL_i(l) + RL_{i'}(l'))/2) \cdot \text{Min}(RL_i(l), RL_{i'}(l'))] \quad (1)$$

In Eq. (1), $(RL_i(l) + RL_{i'}(l'))/2$ is the average of RLs' values on Area i through the two different tasks, Tasks l and l' assigned at Workstation j . The average value is either intensified or diminished by a multiplication of $\text{Min}(RL_i(l), RL_{i'}(l'))$ in order to influence the level

of RLs on the result. So, high RLs increase the product value, dramatically, whereas the lower ones can make the value less or even zero. ARP_j^i presents the level of accumulated risk of posture only on Area i at Workstation j through the assigned tasks at the workstation, and their RLs related to the body area.

On the other side, body areas have their own range of codes to assessment. For example, in Fig. 1c, there are seven codes, and consequently, seven risk levels for leg, whereas these levels for arms are limited to only three codes. Therefore, Coefficients λ_i s are considered to adjust the ranges even based on the level of importance of each body area in the studying AL. However, these coefficients should be set through an iterative decision making. Eq. (2) presents the maximal value of the accumulated risk of postures among all considering body areas in Workstation j .

$$ARP_j = \text{Max}\{\lambda_i \cdot [ARP_j^i]\}_{1 \leq i \leq m} \quad (2)$$

Eventually, ARP in Eq. (3) is presented. The maximal value of ARP_j s gained from all workstations is a new criterion for the proposing ALB problem that should be regarded and minimized.

$$ARP = \text{Max}\{ARP_j\}_{1 \leq j \leq NW} \quad (3)$$

The rest of this subsection deals with an illustrative example to show the ARP calculation and its interpretation.

Example. An AL consists of 21 assembly tasks and five workstations. All necessary information, including task times, overall workload and RLs for two body's areas, A and B, for each task, are reported in Table 2. Simply, overall workload of each task is set to an integer number between 1 and 10, and there are no precedence limitations among these tasks. RL values are also integer values between 0 and 4 meaning that 0 indicates minimal risk, and four expresses maximal risk in posture emerged during the performing of tasks. In this case, the maximal RL that observed in the line is 3.

Five configurations for the ALB problem are chosen and shown in Table 3. The number that represented for configurations in Table 3 indicates the workstation number which the corresponding task is assigned to.

The values of ARP_j^A and ARP_j^B (on Areas A and B, respectively) at Workstation j are calculated by Eq. (1) and presented for each configuration in Table 4 ($1 \leq j \leq 5$, $1 \leq i \leq 2$). Coefficients λ_1 and λ_2 are set to an equal numbers (for instance, 1) due to the same value ranges considering for RL on Area A and B. Therefore, the maximal

Table 2

Details of tasks and their criteria values.

Task #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Time	4	3	9	5	9	4	8	7	5	1	3	1	5	3	5	3	13	5	2	3	7
Overall workload	4	5	2	7	9	10	5	2	10	7	7	2	8	7	7	4	10	10	10	5	6
RL on A	2	1	2	1	3	3	2	1	1	3	3	3	2	1	1	2	1	1	2	0	3
RL on B	0	1	1	2	1	3	3	3	2	1	2	2	0	1	1	1	1	2	2	0	2

Note: RLs' values $\in \{0, 1, 2, 3, 4\}$, 1: minimal value, 4: maximal value.**Table 3**

Five configurations for the ALB problem presented in Table 2 (without precedence limitations).

Task #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Conf. 1	3	4	4	3	4	2	5	2	2	3	3	5	3	1	2	3	1	5	1	1	5
Conf. 2	3	5	1	4	5	1	3	4	3	2	5	1	5	1	1	4	2	3	4	4	2
Conf. 3	2	3	4	3	5	2	2	3	2	4	1	4	1	5	4	3	1	4	3	5	5
Conf. 4	2	5	1	5	2	3	5	1	2	4	1	4	5	4	3	4	4	3	1	2	3
Conf. 5	1	2	2	3	1	3	4	5	1	5	4	5	2	3	5	4	4	3	1	5	2

Table 4Total time in workstations, RLs and ARP_j^i values for the five configurations presented in Table 3.

Workstations	Total Time	RLs on A	ARP_j^A	RLs on B	ARP_j^B
<i>Conf. 1</i>					
Workstation 1	21	0, 1, 1, 2	4	0, 1, 1, 2	4
Workstation 2	21	1, 1, 1, 3	9	1, 2, 3, 3	24.5
Workstation 3	21	1, 2, 2, 3, 3	40	0, 0, 1, 1, 2, 2	11
Workstation 4	21	1, 2, 3	8.5	1, 1, 1	3
Workstation 5	21	1, 2, 3, 3	24.5	2, 2, 2, 3	25
<i>Conf. 2</i>					
Workstation 1	22	1, 1, 2, 3, 3	31	1, 1, 1, 2, 3	18.5
Workstation 2	21	1, 3, 3	13	1, 1, 2	4
Workstation 3	17	1, 2, 2	7	0, 2, 3	5
Workstation 4	20	0, 1, 1, 2, 2	11	0, 1, 2, 2, 3	19
Workstation 5	25	1, 1, 2, 3, 3	31	0, 1, 1, 2, 2	11
<i>Conf. 3</i>					
Workstation 1	21	1, 2, 3	8.5	0, 1, 2	1.5
Workstation 2	21	1, 2, 2, 3	19	0, 2, 3, 3	19
Workstation 3	20	1, 1, 1, 2, 2	17	1, 1, 2, 2, 3	25
Workstation 4	21	1, 1, 2, 3, 3	31	1, 1, 1, 2, 2	17
Workstation 5	22	0, 1, 3, 3	13	0, 1, 1, 2	4
<i>Conf. 4</i>					
Workstation 1	21	1, 2, 2, 3	19	1, 2, 2, 3	19
Workstation 2	21	0, 1, 2, 3	8.5	0, 0, 1, 2	1.5
Workstation 3	21	1, 1, 3, 3	13	1, 2, 2, 3	19
Workstation 4	21	1, 1, 2, 3, 3	31	1, 1, 1, 1, 2	12
Workstation 5	21	1, 1, 2, 2	11	0, 1, 2, 3	8.5
<i>Conf. 5</i>					
Workstation 1	20	1, 2, 2, 3	21	0, 1, 2, 2	7
Workstation 2	24	1, 1, 1, 3	9	0, 1, 1, 2	4
Workstation 3	17	1, 2, 2, 3	19	1, 2, 2, 3	19
Workstation 4	27	1, 2, 2, 3	19	1, 1, 2, 3	13
Workstation 5	17	0, 1, 1, 3, 3	18	0, 1, 1, 2, 3	13

Note: $1 \leq j \leq 5$ $\lambda_1 = \lambda_2 = 1$.

Bold values: The maximal total time among workstations as cycle time.

Italicized values: The Maximal ARP_j^i as ARP_j , and also as ARP .

value of these ARP_j^i 's can be simply indicated as ARP_j and also ARP for each configuration. These values are shown by bold rectangles in the table.

Conf. 5 offers less maximum value of ARP , 21, whereas this value is 40, 31, 31 and 31 for Conf. 1, 2, 3 and 4 respectively. Obviously, the highest value of RLs, 3, is almost uniformly distributed among all workstations in Conf. 5. Although, Conf. 5 gives a nearly good diversity in high-risk postures, it results a cycle time of 27 units of time, and this time is 21 for Conf. 1 with ARP of 40 and also for Conf. 4 with ARP of 31. According to total time and ARP , Conf. 4 has a minimal cycle time of 21; and it yields a maximum ARP value

of 31. Apparently, this configuration can be such a compromise solution for ALB problem, but with regard of only two criteria: cycle time and ARP . However, the decision could be more sophisticated as it will be mentioned in the coming sections.

3.2. Mathematical formulation

The proposing ALB model is concerned with three criteria; cycle time, overall physical workload and ARP . The notations used in the model, and its objectives are introduced as follows:

$$\text{Minimizing } f_1(X) = \text{Max} \left\{ \sum_{i \in T_j} t_i - \mu \right\}_{1 \leq j \leq NW} \quad (4)$$

$$\text{Minimizing } f_2(X) = \text{Max} \left\{ \sum_{i \in T_j} OW_i - \rho \right\}_{1 \leq j \leq NW} \quad (5)$$

$$\text{Minimizing } f_3(X) = ARP \quad (6)$$

Subject to

$$X_j \leq X_i, \quad \forall i, j \in P_i \quad (7)$$

$$X_i \in \{1, \dots, NW\} \quad (8)$$

Eq. (4) describes the first well-known objective of ALB problem that tries to balance the cycle times among all workstations. It happens through the minimizing of maximal deviation between total processing times ($\sum_{i \in T_j} t_i$) in workstations and the ideal cycle time (μ). The second objective presented in Eq. (5), works to balance the overall physical workload among workstations regarding the average of the workload in entire the line. The effort is to minimize the maximal deviation between summation of the workloads in each workstation j ($\sum_{i \in T_j} OW_i$) and its average (ρ). The third objective that is the main contribution of this research is formulated by Eq. (6). It minimizes the maximal value of $ARPs$ on considering body areas over the all workstations. Besides that, Constraint (7) shows the precedence limitation, as a conventional constraint in ALB problem; and Constraint (8) means that each task must be assigned to only one workstation (Tasan & Tunali, 2008).

3.3. Fuzzy goal programming

In goal programming, which is probably the original multiple-criteria decision making technique, the decision maker sets

desirable goals for somewhat conflicting objectives. In many cases, it may be difficult to define imprecise or uncertain goals (Ozcan & Toklu, 2009). Therefore, fuzzy sets represent an applicable tool for dealing with this situation (Karwowski, Lee, & Cho, 1998). Fuzzy goal programming was initially developed from fuzzy environments (Narasimhan, 1980).

This research considers two fuzzy goals. The first fuzzy goal is the cycle time that always is a part of all ALB problems. The second fuzzy goal, the overall physical workload standard that can be defined through some previous works like Choi's model (2009).

Non-preemptive and preemptive goal programming can be expanded to accommodate fuzzy goal programming. In non-preemptive programming, all goals have been considered to have roughly equal importance; in the preemptive approach, the goals are prioritized in a hierarchy (Gen & Cheng, 2000). This study uses a non-preemptive approach, as shown in the following steps.

3.3.1. Step 1: define goals

As mentioned above, two goals are considered and defined as follows:

Goal 1 – The maximal deviation of cycle time for the assembly line, z_1 should not exceed CT , as it is formulated by Eq. (9).

$$z_1 = \text{Max} \left\{ \sum_{i \in T_j} t_i - \mu \right\}_{1 \leq j \leq NW} < CT \quad (9)$$

Goal 2 – The maximal deviation of overall workload at workstations, z_2 should not exceed OW , as it is formulated by Eq. (10).

$$z_2 = \text{Max} \left\{ \sum_{i \in T_j} OW_i - \rho \right\}_{1 \leq j \leq NW} < OW \quad (10)$$

3.3.2. Step 2: defend membership function

We consider t and w as upper tolerance limits; and CT and OW as lower tolerance limits to the fuzzy goals, z_1 and z_2 respectively. However, rough values of these limits can be gained through an iterative decision making. The related membership functions are illustrated in Fig. 2.

The linear membership functions for the fuzzy goals, μ_{z_1} and μ_{z_2} , are defined as follows:

$$\mu_{z_1} = \begin{cases} 1 & \text{if } z_1 \leq CT \\ \frac{t-z_1}{t-CT} & \text{if } CT < z_1 < t \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

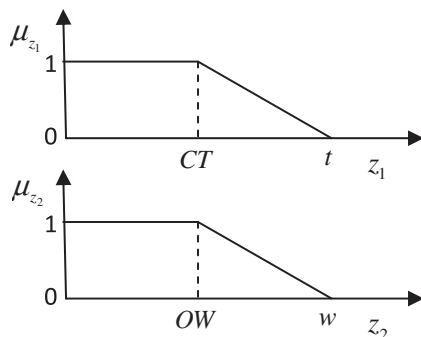


Fig. 2. Membership functions of the fuzzy goals representing the positive deviation of cycle time (up) and overall workload (bottom).

$$\mu_{z_1} = \begin{cases} 1 & \text{if } z_2 \leq OW \\ \frac{w-z_2}{w-OW} & \text{if } OW < z_2 < w \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

The fuzzy goals are then converted to the following form:

$$\mu_{z_1} \leq \frac{t-z_1}{t-CT} \quad (13)$$

$$\mu_{z_2} \leq \frac{w-z_2}{w-OW} \quad (14)$$

$$0 \leq \mu_{z_1}, \mu_{z_2} \leq 1 \quad (15)$$

Selecting a specified goal for third objective, f_3 , obviously depends on the number of workstations and involving tasks in each workstation. As it is mentioned earliest, the third objective tries to minimize the possibly maximal value of ARP resulted at workstations, not balancing it. So, determination of the average of ARP as a safe or desirable goal is not the point here.

Eventually, the proposing ALB model with its objective function is presented as follows:

$$\text{Maximizing} \left[w_1 \cdot \mu_{z_1} + w_2 \cdot \mu_{z_2} + \frac{w_3}{f_3(X)} \right] \quad (16)$$

Subject to fuzzy goal constraints:

$$(t-CT) \cdot \mu_{z_1} + \text{Max} \left\{ \sum_{i \in T_j} t_i - \mu \right\}_{1 \leq j \leq NW} - t \leq 0 \quad (17)$$

$$(w-OW) \cdot \mu_{z_2} + \text{Max} \left\{ \sum_{i \in T_j} OW_i - \rho \right\}_{1 \leq j \leq NW} - w \leq 0 \quad (18)$$

and two conventional constraints of ALB problem presented by (7) and (8).

Objective presented in (16), shows an aggregated objective function that is a mixed fuzzy and non-fuzzy decision with the weighted achievement degree of the fuzzy goals and the non-fuzzy objective. In addition to the conventional constraints in any ALB problem, Constraints (7) and (8), Constraints (17) and (18) are considered to regard the lower and upper limits of the two fuzzy goals.

In the proposed model, a set of desirable levels of the achievement degree of the fuzzy goal can be also determined as constraints, i.e., $\mu_{z_i} \geq \delta_i$, where $\delta_i (0 \leq \delta_i \leq 1)$ is the desirable achievement degree for the i th fuzzy goal. Although, the determination is a difficult task for the decision makers(s), some useful methods can be used to set a fuzzy goal importance as a real number in the range of [0, 1] (Chen & Tsai, 2001). Moreover, the constraints cannot be considered as fuzzy goals, because a compatible or an incompatible zoning can be either satisfied or unsatisfied.

4. Genetic algorithm

Genetic Algorithms (GAs) are interactive search approaches based on the selection natural process and inheritance across generations. GAs maintain a population of some (mostly fixed number of) candidate members across generations and aims to select better ones based on their fitness values for reproduction. Generally, the main steps of GA are as follows:

- Step (1): Creating initial population (often randomly generated).
- Step (2): Evaluating the population members (for fitness).
- Step (3): Selecting two individuals as parents to undergo two main genetic operations (crossover and mutation).
- Step (4): Evaluating the new individuals (as offspring).

- Step (5): Applying replacements and make a new population from the old population and new offspring.
- Step (6): If the stop conditions are met, then stop, otherwise, return to step 3.

Depending on the problem to be solved by GAs, GA parameters and structures can be altered. In (Tasan & Tunali, 2008), the authors have surveyed much recent GAs for different types of ALB problem. In the following subsections, we present some useful explanations about individual representation, population, fitness and genetic operations used in our algorithm.

4.1. Representation

Task-based representation (Sabuncuoglu, Erel, & Tanyer, 2000) and workstation-based representation (Kim, Kim, & Kim, 2000) are two general representation forms. Task-based representation is most appropriate for a type of ALB problems that is used to minimize the number of workstations. Workstation-based representation is generally used for the other type of ALB problem where the minimal cycle time is a main objective. Then, in this paper, workstation-based representation is used as follows.

The chromosome is made into a vector containing the workstations' labels (or indices) that show which tasks are assigned to which workstation. The length of the chromosome is fixed by the number of tasks. For example, Fig. 3 shows an AL with seven tasks, three workstations and a related precedence graph (see the direction in the graph that means the precedence). The workstation-based chromosome presented in the figure is: (1, 2, 1, 2, 3, 3, 3) that it means Tasks 1, 3 are assigned to Workstation 1; Task 2 and 4 are assigned to Workstation 2; and finally, Task 5, 6 and 7 are assigned to Workstation 3.

It is noteworthy that no precedence relation has been violated in the representation. To keep this, all genetic operations should be designed by a good strategy to avoid destructing the precedence.

4.2. Initial population and fitness function

The initial population is composed of a set of chromosomes (called individuals) that are generated randomly (with no violation in precedence limitations). The initially generated population might not ensure a lack of convergence into the local optimums. So, in this paper, we divided the population into two sections (30–70) to improve the evolutionary search. A larger part of the population (70 from 100) constantly belongs to new generation from previous population, and the rest (30 from 100) are seized by the individuals randomly generated each generation (exactly like initial individuals). In fact, the population always contains fresh individuals that can keep the evolutionary search away from lasting in local optima for long.

The value of the fitness function is the same as function F , presented in (14), but the weights must be set to real values by the decision maker(s). An interactive approach can be taken to adapt the weights or the other useful methods can be employed as it is

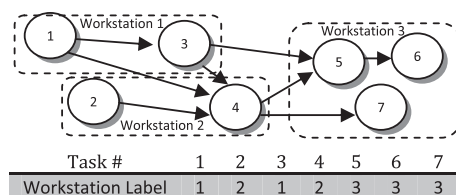


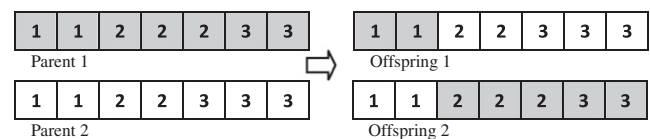
Fig. 3. An one workstation-based chromosome representation.

mentioned in at the end of Section 2. We set them to 1 in order to give them the same level of importance.

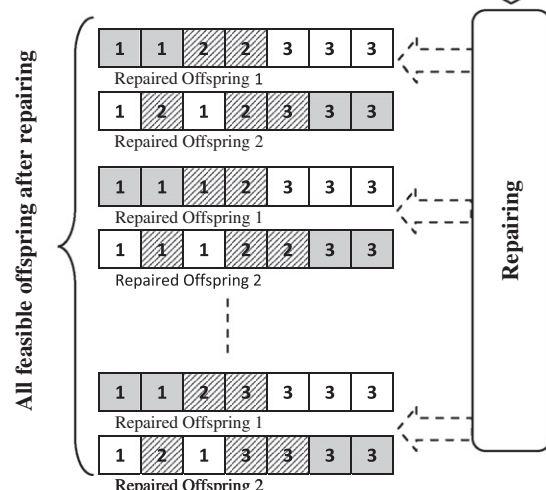
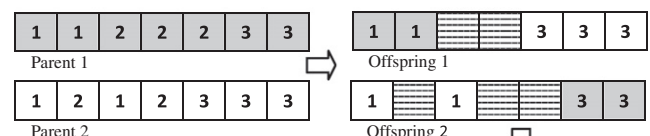
4.3. Genetic operations

The main genetic operation is the crossover, which combines pieces of information from selected individuals by the selection operation. Simaria and Vilarinho (2004) introduced a crossover mechanism that ensures feasibility without breaking precedence limitations. The single crossover point (CP) is a randomly generated integer between 1 and NT (number of tasks). To generate the first offspring, the assignment of Workstations 1 to CP is copied from the first parent, and the remaining positions are copied from the assignment of Workstation CP + 1 to NT from the second parent. For the second offspring, the process must be repeated after exchanging the first parent with the second.

For instance, Fig. 4 illustrates two solutions from the example presented in Fig. 3, and shows how a crossover would be performed. In Fig. 4a, CP is equal to 1, and the crossover operation can be done easily. In Fig. 4b, it is a bit more complicated. There are no empty positions to exchange; therefore, an overhead operation, "Repairing," is required. Repairing operation surveys all feasible workstation labels that can fill the remaining empty positions according to precedence limitations. As shown in Fig. 4b, empty positions are marked by striped lines. With Repairing, all feasible labels of workstations are used to fill them. Finally, repaired solutions that have minimum idle times in workstations are selected for next operation, mutation.



a) Crossover operation with CP=1 / no need to repair



b) Crossover operation with CP=1 / need to repair with feasible choices

Fig. 4. Crossover operation and repairing operation.

A mutation operation is performed with a low level of probability to allow a stochastic search only by changing a small amount of information of the individuals. An individual is selected to undergo mutation according to the small mutation probability, and a task is randomly selected. The workstation label assigned to the selected task must be changed to a label between the maximum amount of labels of workstations assigned to the preceding tasks of the selected task in the precedence graph, and the minimum amount of labels of workstation assigned to succeeding tasks of the selected task in the precedence graph.

5. Illustrative example

In this section, some comparison results of the example given in Section 3 are presented. First, for the fuzzy goals (z_1 and z_2), the lower tolerance limits (CT and OW) and the upper tolerance limits (t and w) are simply set to 0, 0, 10 and 20 respectively. Although the weights in Eq. (16) can be set in an interactive way, these are simply set to 1.

In order to make different configurations for more study, various combinations of the studied criteria are considered. Therefore,

Table 5

Ten different configurations for ALB problem with five workstations presented in Table 2.

Task #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Conf. 1	3	4	4	3	4	2	5	2	2	3	3	5	3	1	2	3	1	5	1	1	5
Conf. 2	3	5	1	4	5	1	3	4	3	2	5	1	5	1	1	4	2	3	4	4	2
Conf. 3	2	3	4	3	5	2	2	3	2	4	1	4	1	5	4	3	1	4	3	5	5
Conf. 4	2	5	1	5	2	3	5	1	2	4	1	4	5	4	3	4	4	3	1	2	3
Conf. 5	1	2	2	3	1	3	4	5	1	5	4	5	2	3	5	4	4	3	1	5	2
Conf. 6	1	1	1	1	2	3	3	3	4	4	4	4	4	3	4	4	5	5	5	5	2
Conf. 7	1	1	1	1	2	2	3	3	3	4	4	4	4	3	4	4	5	5	5	5	2
Conf. 8	1	1	1	1	2	2	3	3	3	3	4	3	4	4	4	4	5	5	5	5	2
Conf. 9	1	1	1	1	2	2	3	3	3	3	4	5	4	4	4	5	5	4	5	5	2
Conf. 10	1	1	1	1	2	3	2	3	3	4	4	5	3	3	4	5	5	5	5	5	2

Table 6

Ten solutions as new configurations for the assembly line with/without considering precedence limitations.

With precedence limitation					Without precedence limitation				
	Proc. time	Overall workload	LRs on A	LRs On B		Proc. time	Overall workload	LRs on A	LRs On B
<i>Details of Conf. 1 with precedence limitation</i>					<i>Details of Conf. 6 without precedence limitation</i>				
Regarding CTD criterion					Regarding CTD criterion				
WS 1	21	18	1, 1, 2, 2	0, 1, 1, 2	WS 1	21	32	0, 1, 1, 2	0, 1, 1, 2
WS 2	16	15	3, 3	1, 2	WS 2	21	29	1, 1, 1, 3	1, 2, 3, 3
WS 3	22	24	1, 1, 2, 3	1, 3, 3, 3	WS 3	21	37	1, 2, 2, 3, 3	0, 0, 1, 1, 2, 2
WS 4	23	45	1, 1, 2, 2, 3, 3, 3	0, 1, 1, 1, 2, 2, 2	WS 4	21	16	1, 2, 3	1, 1, 1
WS 5	23	35	0, 1, 1, 2	0, 1, 2, 2	WS 5	21	23	1, 2, 3, 3	2, 2, 2, 3
<i>Details of Conf. 2 with precedence limitation</i>					<i>Details of Conf. 7 without precedence limitation</i>				
Regarding CTD and OWD Criteria					Regarding CTD and OWD Criteria				
WS 1	21	18	1, 1, 2, 2	0, 1, 1, 2	WS 1	22	28	1, 1, 2, 3, 3	1, 1, 1, 2, 3
WS 2	20	25	3, 3, 3	1, 2, 3	WS 2	21	23	1, 3, 3	1, 1, 2
WS 3	23	24	1, 1, 1, 2	1, 2, 3, 3	WS 3	17	19	0, 1, 2	0, 2, 3
WS 4	18	35	1, 2, 2, 3, 3, 3	0, 1, 1, 1, 2, 2	WS 4	20	28	0, 1, 1, 2, 2	0, 1, 2, 2, 3
WS 5	23	35	0, 1, 1, 2	0, 1, 2, 2	WS 5	25	39	1, 1, 2, 3, 3	0, 1, 1, 2, 2
<i>Details of Conf. 3 with precedence limitation</i>					<i>Details of Conf. 8 without precedence limitation</i>				
Regarding CTD and OWD and ARPs criteria					Regarding CTD and OWD and ARPs criteria				
WS 1	21	18	1, 1, 2, 2	0, 1, 1, 2	WS 1	21	25	1, 2, 3	0, 1, 2
WS 2	20	25	3, 3, 3	1, 2, 3	WS 2	21	29	1, 2, 2, 3	0, 2, 3, 3
WS 3	22	26	1, 1, 2, 3, 3	1, 2, 2, 3, 3	WS 3	20	28	1, 1, 1, 2, 2	1, 1, 2, 2, 3
WS 4	19	33	1, 1, 2, 2, 3	0, 1, 1, 1, 2	WS 4	21	28	1, 1, 2, 3, 3	1, 1, 1, 2, 2
WS 5	23	35	0, 1, 1, 2	0, 1, 2, 2	WS 5	22	27	0, 1, 3, 3	0, 1, 1, 2
<i>Details of Conf. 4 with precedence limitation</i>					<i>Details of Conf. 9 without precedence limitation</i>				
Regarding CTD and ARPs criteria					Regarding CTD and ARPs criteria				
WS 1	21	18	1, 1, 2, 2	0, 1, 1, 2	WS 1	21	21	1, 2, 2, 3	1, 2, 2, 3
WS 2	20	25	3, 3, 3	1, 2, 3	WS 2	21	28	0, 1, 2, 3	0, 0, 1, 2
WS 3	21	24	1, 1, 2, 3	1, 2, 3, 3	WS 3	21	33	1, 1, 3, 3	1, 2, 2, 3
WS 4	21	39	1, 1, 1, 2, 3	0, 1, 1, 2, 2	WS 4	21	30	1, 1, 2, 3, 3	1, 1, 1, 1, 2
WS 5	22	31	0, 1, 2, 2, 3	0, 1, 1, 2, 2	WS 5	21	25	1, 1, 2, 2	0, 1, 2, 3
<i>Details of Conf. 5 with precedence limitation</i>					<i>Details of Conf. 10 without precedence limitation</i>				
Regarding ARPs criterion					Regarding ARPs criterion				
WS 1	21	18	1, 1, 2, 2	0, 1, 1, 2	WS 1	20	33	1, 2, 2, 3	0, 1, 2, 2
WS 2	24	20	2, 3, 3	1, 2, 3	WS 2	24	21	1, 1, 1, 3	0, 1, 1, 2
WS 3	24	37	1, 1, 1, 2, 3	0, 1, 2, 3, 3	WS 3	17	34	1, 2, 2, 3	1, 2, 2, 3
WS 4	9	21	1, 3, 3	1, 1, 2	WS 4	27	26	1, 2, 2, 3	1, 1, 2, 3
WS 5	27	41	0, 1, 1, 2, 2, 3	0, 1, 1, 2, 2, 2	WS 5	17	23	0, 1, 1, 3, 3	0, 1, 1, 2, 3

Note: The italicized values shows the workstation with high RLs (3 or higher) on the related the body's areas.

deviation of cycle time (CTD); CTD and positive deviation of overall workload (OWD); CTD, OWD and ARP; CTD and ARP; and ARP alone, are selected as five combinations.

Additionally, we use a GA in two different conditions of with and without of precedence limitations. The only reason of making these conditions is just to study of the influence of the limitations on the results. The population size, crossover and mutation rates are also set to 50, 0.9 and 0.1 respectively. The best solution or configuration for the ALB example can be released from the none-dominated solutions in the final population in GA.

According to the above considerations, two general groups of configurations (solutions) can be gained (see Table 5). First group comprises of five different configurations: Conf. 1, Conf. 2, Conf. 3, Conf. 4 and Conf. 5. There have been founded by GA regarding the five combinations of criteria with no precedence limitations. And Conf. 6, Conf. 7, Conf. 8, Conf. 9 and Conf. 10, as the second group, have been resulted by GA regarding the same combination of criteria except precedence limitation; that it has been considered for this group.

The details of all 10 configurations are presented in Table 6. Five configurations with precedence limitations are shown on the left side and five configurations without precedence limitations shown on the right side. All situations that include at least one high-risk posture (value of 3 or 4) are bounded with bold rectangles in the table.

Supposing that in a workstation with more than two 3s or higher for RL means much relatively pressure on the related body area, there is no such as this pressure on body areas in Conf. 5 to Conf. 10. In Conf. 10, where the only concern is ARP disregarding the precedence limitations, a good distribution of high-risk postures (as high-risk posture diversity) is attained. Only one workstation, WS 5, is assigned with two high RLs (two 3s in A). However, its cycle time (is equal to 27) and the overall physical workload (is equal to 34) are not better than others.

This example demonstrates the somewhat conflicting criteria, process time (or cycle time as maximal process time), overall workload and ARP. However, these should be considered simultaneously to obtain a compromise solution for ALB problem. Although the fuzzy goals in the proposed ALM model make the way of seeking of configurations much flexible, in real applications, some criteria cannot be easy to determine by decision makers. Therefore, using an interactive approach or having more solutions (for example, Pareto solutions) is important for the decision making.

6. Computational experimental results

Scholl's benchmark data (Scholl, 1999), which give dozen ALB problems with different precedence limitations and processing times, were used. For additional data, a 10-point scale variable set were selected to generate overall workloads, and a 5-point scale variable set (from 0 to 4) was considered for RL's values on only two areas 'A' and 'B'. However, the number of body areas can increase. The values of overall workload were randomly generated by a uniform distribution function, but the RL's values were generated by follows the Eq. (19).

$$\forall R: \text{Random number between } 0 \text{ and } 2^4$$

$$RL_{1 \text{ or } 2}(i) = k, \quad \text{if } 2^{k-1} \leq R < 2^k, \quad \forall i \in 1, 2, \dots, NT \quad (19)$$

The exponential distribution in (19) makes the generation of higher-risk levels less likely. Because it is assumed that only some assembly tasks (not all) demand awkward postures or high-risk RL values, otherwise the contributions proposed in this paper would be nearly worthless. And the assembly line needs to re-design in terms of ergonomics issue, indeed.

Table 7

The all information of 17 ALB problems and the gained values for CT, OW and ARPs.

ALB problems	Number of tasks	Number of workstations	Optimal positive deviation of CT	Best found solution regarding only CTD				Best found solution regarding only CTD and OWD				Best found solution regarding CTD and OWD and Max. iteration			
				Gained positive deviation of OW	Gained positive deviation of CT	Max. of ARP ^A	Max. of ARP ^B	Gained positive deviation of OW	Gained positive deviation of CT	Max. of ARP ^A	Max. of ARP ^B	Gained positive deviation of OW	Gained positive deviation of CT	Max. of ARP ^A	Max. of ARP ^B
Buxey	29	14	1.857	1.8571	8.1429	4	9	1.8571	1.8571	7.5	7.5	6.1429	1.8571	4	7
		13	2.077	2.077	11.3077	7.5	8.5	3.0777	3.0777	4	14	5.3077	2.0777	7	8.5
		12	1	3	10.3333	10	19	6.3333	3	11	14	4.3333	3	7.5	9
		10	1.6	1.6	7.8	11	14	4.8	1.6	11	14	3.8	1.6	10	13
		7	0.714	2.7143	7.2857	28	33.5	3.2857	2.7143	16.5	49.5	3.2857	2.7143	16.5	27
Kilbridge	45	11	4.8182	4.8182	16.2727	62	28	5.2727	4.8182	30	25	4.2727	4.8182	22	18.5
		10	0.8	1.8	24.3	52	52	6.3	1.8	45	34.5	5.3	2.8	21.5	21.5
		8	0	1	10.875	76	85.5	5.875	3	54	89	5.875	2	37	44
		6	0	1	15.8333	75	113	5.8333	1	122.5	126.5	5.8333	1	49.5	47
		4	0	1	17.75	114	126	1.75	1	152.5	152.5	0.75	1	114	139.5
Lutz2	89	3	0	0	9.6667	206	206	1.6667	0	247	247	0.6667	1	152	159.5
		28	0.679	2.6786	21.8571	27	34.5	8.8571	3.6786	42	31	7.857	3.678	27.5	27
		25	0.6	2.6	19.68	42.5	71	11.68	1.6	27.5	59.5	5.68	2.6	15	18
		20	0.75	1.75	11.6	72	89.5	9.6	1.75	52	51	6.6	1.75	34	31
		15	0.667	1.6667	19.1333	100.5	106	5.1333	2.6667	82.5	86	5.1333	2.666	44	47
		10	0.5	1.5	26.2	150.5	146.5	6.2	1.5	195.5	182	6.2	1.5	108	106
		9	0.111	1.1111	19.5556	229.5	238	6.5556	1.1111	195.5	203	3.5556	2.11	123.5	146.5

Note: j is an index used for workstation number.

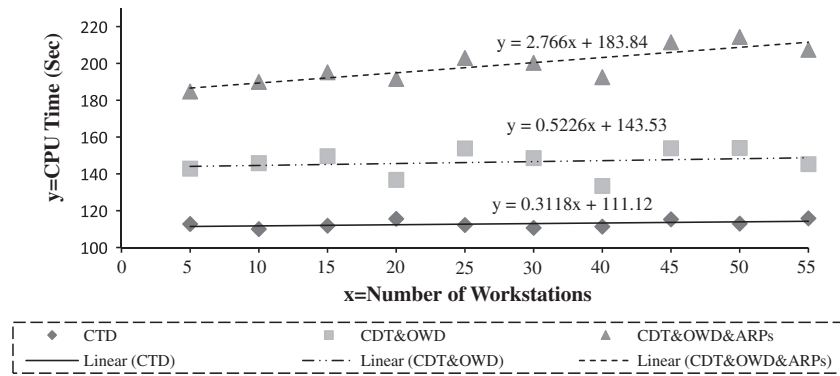


Fig. 5. CPU time spending analysis for Lutz2 Problem (after 200 iterations) regarding the three models (CTD, CTD and OWD and CTD and OWD and ARPs).

Afterward, three well-known ALB problems, Buxey, Kilbridge and Lutz-2 from Scholl's benchmark set (Scholl, 1999), were extended with the two new parameters. And with the given numbers of workstations, 17 ALB problems were generated with the 10-point and 5-point scale variables for overall workloads and LRs. All the test cases and the related GAs were executed on the computer with Intel® Core™ Quad CPU Q9550@2.83 GHz; and also, MathWork Matlab® software package R2009a was used to implement the GAs.

Three combinations of the criteria; CTD, CTD and OWD, and CTD and OWD and ARP were studied, and their ALB models were evaluated. Their best solutions found by fixed iterations, are shown in Table 7. Buxey with 14 and 10 workstations and Kilbridge with 11 workstations, have found their own optimal CTD (Deviation of CT) in all three models (1.857 and 1.6 for Buxey with 14 and 10 workstation, respectively, and 4.8182 for Kilbridge with 11 workstations).

Due to evolutionary search of GAs, it is not always expected to meet the similar results for one problem after the same number of iterations. In Buxey case with 13 workstations, the second model regarding CTD and OWD found the value of 3.0769 for CTD whereas the third model regarding more criteria, CTD and OWD and ARPs, reached the smaller value of 2.0769 for CTD. Actually, these values are only gained by approximations through evolutions in their GAs. These values could change as to whether the evolutions go forward. The same happened to Kilbridge with 8 workstations and Lutz2 with 25 workstations.

To show the CPU times spent by the three models, Fig. 5 illustrates the growth of CPU consuming while the number of workstations increases up to 55 for Lutz-2 case. The related linear regressions are also presented in the diagram. As it is expected, the GA dealing with third model, ALB regarding CTD and OWD and ARP, spent more CPU than others as the number of workstations increase.

7. Conclusion and future research directions

In this research, we considered high risks of body postures to workers who perform assembly tasks in workstations. To end this, a new parameter, Accumulated Risk of Postures (ARPs) was introduced to estimate levels of risks on a specified area of workers' body in a workstation. We defined it so that its value reflects a lack of posture changes or a level of monotonous postures (high-risk ones) during workers' job in a workstation. Afterward, ARP that contributed to ALB models should be minimized in each workstation along other objectives. In the proposed model, we considered ARP as a supplementary criterion to overall physical workload and cycle time. Three somewhat conflicting objectives were considered simultaneously to obtain effective and realistic configurations of

ALs. Moreover, to find the solutions, we defined two goals, positive deviation of cycle time (CTD) and positive deviation of overall physical workload (OWD). The goals cannot be easily determined as precise goals by the decision, and then we used fuzzy goals and fuzzy goal programming to formulate the model. With the combinations of the criteria and their objectives, five models were created. Through an illustrated example and computational experimental tests, the expected results showed how well ARP influence ALB. However, we do not emphasis on ARP as a primary criterion for ALB problem and it should be regarded as compromise decision(s).

We tried to make a simple ARP definition with a simplified version of OWAS to code the risk of postures involved. As a further work, other exposure assessments such as REBA, RULA and QEC, which give an individual score for each body area can be also used as the assessments benefit from posture-based observation.

The proposed definition and ARP objective can also be used to enhance other types of ALs: mixed-model, U-shape and parallel two-side.

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